Autoignition Delay Time Measurements of Methane, Ethane, and Propane Pure Fuels and Methane-Based Fuel Blends

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Autoignition delay experiments in air have been performed in an atmospheric flow reactor using typical natural gas components, namely, methane, ethane, and propane. Autoignition delay measurements were also made for binary fuel mixtures of methane/ethane and methane/propane, and ternary mixtures of methane/ethane/propane. The effect of CO_2 addition to the methane-based fuel blends on autoignition delay times was also investigated. Equivalence ratios for the experiments ranged between 0.5 and 1.25, and temperatures ranged from 930 K to 1140 K. Consistent with past studies, increasing equivalence ratio and increasing inlet temperatures over these ranges decreased autoignition delay times. Furthermore, addition of 5–10% ethane or propane decreased autoignition delay time of the binary methane-based fuel by 30–50%. Further addition of either ethane or propane showed less significant reduction of autoignition delays. Addition of 5–10% CO_2 slightly decreased the autoignition delay times of methane fuel mixtures. Arrhenius correlations were used to derive activation energies for the ignition of the pure fuels and their mixtures. Results show a reduction in activation energies at the higher temperatures studied, which suggests a change in ignition chemistry at very high temperatures. Measurements show relatively good agreement with predictions from a detailed kinetics mechanism, specifically developed to model ignition chemistry of C1-C3 alkanes. [DOI: 10.1115/1.4000590]

1 Introduction

Natural gas is a common fuel used in gas turbines for power generation [1]. It is becoming increasingly advantageous for industrial gas turbines to be capable of operating on a variety of fuel blends while maintaining performance criteria such as low emissions and flame stability [2,3]. The fuels of interest are liquefied natural gases from all over the world, coal-derived syngas, biomass, landfill gas, and process gas. These fuels have a broad range of compositions, which were shown to have significant effects on combustor performance, affecting efficiency, stability, and emissions [2]. Natural gas composition varies globally and, to a lesser extent, within the United States. U.S. natural gas composition can contain methane concentrations as low as 74.5% (by volume) and ethane and propane concentrations as high as 13.3% and 23.7% (by volume), respectively, with the balance made up of higher-order hydrocarbons, carbon dioxide, oxygen, nitrogen, and hydrogen sulfide [4–6].

Even a few percent of higher-order hydrocarbon was shown to significantly promote the ignition of methane-based fuels [5]. This is of serious concern in premixed systems such as industrial dry, low emission (DLE) gas turbines, which often use aerodynamic methods to stabilize the flame in a particular zone. Variations in autoignition phenomena, based on fuel composition, could result in flashback and flame stabilization in unintended locations, such as the premixer, causing performance problems and hardware damage [7]. Researchers recognized the need to quantify the promotional effects of higher-order hydrocarbons on natural gas ignition and combustion processes [8]. Autoignition delay measurements for natural gas components and mixtures provide valuable insight into the effects of additives on methane ignition, as well as validation data for chemical kinetics mechanisms for predicting ignition of methane-based fuels.

Recent studies showed that natural gas ignition chemistry can be suitably represented by methane/ethane /propane mixtures [9,10]. The current study was established to provide reliable autoignition delay time measurements of methane/ethane/propane pure fuels and mixtures at relevant gas turbine inlet temperatures. The measurements extend the current database of available ignition data for natural gas fuels and provide much needed validation for the development of kinetics models in this parameter space.

Pure methane oxidation has been extensively studied experimentally for a number of years. A summary of the ignition delay time measurements made up to 1969, mostly focusing on temperatures above 1400 K and pressures near atmospheric, is provided by Spadaccini and Colket [7]. In recent years, the parameter space for methane autoignition was well investigated for pressures up to 300 atm and temperatures ranging from 849 K to 1600 K [5,11–17]. Pure ethane and propane oxidation has been less extensively studied. An excellent summary of experimental ethane autoignition delay time measurements is provided by de Vries et al. [18]. Measurements exist for temperatures ranging from 1218 K to 1860 K and pressures up to 3 atm [18,19] for ethane oxidation, and temperatures ranging from 900 K to 2211 K and pressures up to 30 atm for propane oxidation [14,20–23]. Figure 1 summarizes the autoignition delay times for methane, ethane, and propane pure fuels reported in the literature, showing the range of conditions investigated by various researchers.

de Vries et al. [16] provided a summary of experimental studies to date, which have investigated the effects of higher-order hydrocarbons on methane ignition behavior. Goy et al. [12] studied the effect of significant (15% by volume) additions of ethane and propane on methane autoignition delay for temperatures ranging from 952 K to 1428 K and pressures up to 20 atm. Significant changes in activation energy for the fuel blends were observed,

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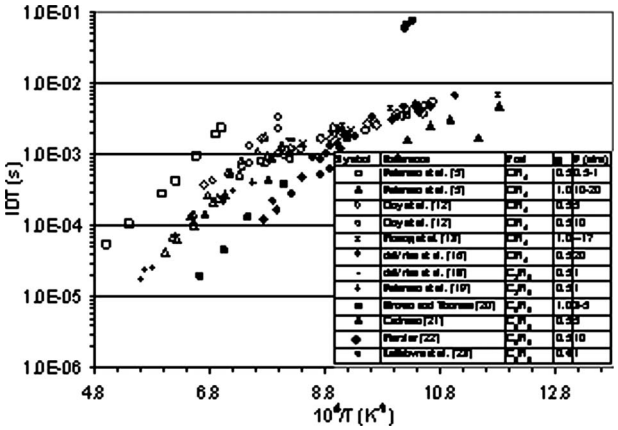

Fig. 1 Summary of autoignition delay times for pure fuels methane, ethane, and propane reported in the literature including experimental conditions

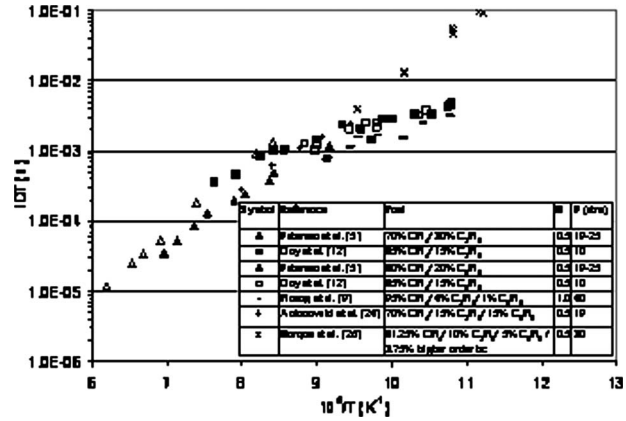

Fig. 2 Summary of autoignition delay times for methane-based fuel blends containing ethane and propane reported in the literature including experimental conditions

which were not captured by available kinetics models. Petersen et al. [1] conducted autoignition delay time experiments for methane/propane fuel blends for pressures up to 25 atm and temperatures ranging from 1091 K to 1655 K, resulting in the modification of some reaction parameters in the GRI 3.0 mechanism. Huang and Bushe [9] studied the effect of ethane and propane on methane autoignition for temperatures ranging from 900 K to 1400 K and high pressures of 16–40 atm. Significant promotional effects of ethane and propane on methane autoignition were observed. Antonovski et al. [24] collected autoignition delay times of several fuel mixtures composed of methane, ethane, and propane for pressure conditions up to 58 atm and temperatures ranging from 1032 K to 1577 K. Antonovski et al. [24] found ethane and propane concentration to have a significant influence on the autoignition of methane. Petersen et al. [5] investigated the effect of significant (>10% by volume) additions of ethane and propane on methane autoignition delay time for temperatures ranging from 1090 K to 2001 K and pressures up to 30 atm. Again, significant reductions in autoignition delay times were observed for the mixtures compared with pure methane measurements. de Vries et al. [16] collected experimental autoignition delay time data for methane blends containing up to 50% ethane and 25% propane for temperatures ranging from 811 K to 1107 K and pressures around 20 atm. Healy et al. [25] continued this research and studied mixtures of methane and propane of ratios 90/10% and 70/30% at temperatures ranging from 740 K to 1550 K and pressures of 10 atm, 20 atm, and 30 atm. The authors observed a negative temperature coefficient for autoignition delay time in the temperature regime studied. A kinetics model developed by Petersen et al. [1] predicted the results of the autoignition delay experiments well, capturing the negative temperature coefficient trend. Bourque et al. [26] continued the work of this group and studied the ignition of natural gas blends of methane, ethane, propane, *n*-butane, and *n*-pentane. The authors reported excellent agreement with the kinetics model [1], especially for fuel-rich mixtures at higher pressures. They recommend further investigation of the effect of higher (up to 40%) concentrations of ethane in the fuel blend, as the model needs further validation in this parameter space. Figure 2 presents selected data sets for methane fuel blends containing ethane and propane reported in the literature, showing the range of conditions investigated and autoignition delay times measured by the various researchers.

Very limited data exist for the atmospheric pressure and temperature range studied herein. Further, limited autoignition delay measurements exist for methane-based blends composed with more than 15% additive. The current study investigates the ignition of methane-based fuel blends containing 0–100% additive (ethane and propane), as well as ternary combinations of methane, ethane, and propane. The collected data spans the equivalence ratio range from 0.5 to 1.25, providing a comprehensive matrix of fuel blends and conditions. This paper presents those measurements beginning with a description of the experimental flow reactor, followed by a discussion on the analysis for determining autoignition delay time. The experimental results are presented with Arrhenius correlations that were derived from the data. The ignition delay time data are also compared with the predictions from plug flow reactor models using a detailed chemical kinetics mechanism.

2 Measurement of Ignition Delay Time

The autoignition delay time measurement apparatus is a modified version of a rig, previously used to measure autoignition delay times of kerosene-type fuels and is described in detail elsewhere [27,28]. A schematic of the experimental system is provided in Fig. 3. The flow reactor apparatus consisted of a fuel preheating system, an air preheating system, a premixing section, and a test section. Preheated fuel was injected into a hot air stream, and after rapid mixing, the fuel/air mixture ignited at some distance down the test section of the flow reactor. The time was recorded by the data acquisition system when the valve controlling fuel injection was opened and referenced as the fuel injection time. The time at which chemiluminescence emissions (excitation of CH^* radicals) were detected using a photomultiplier tube

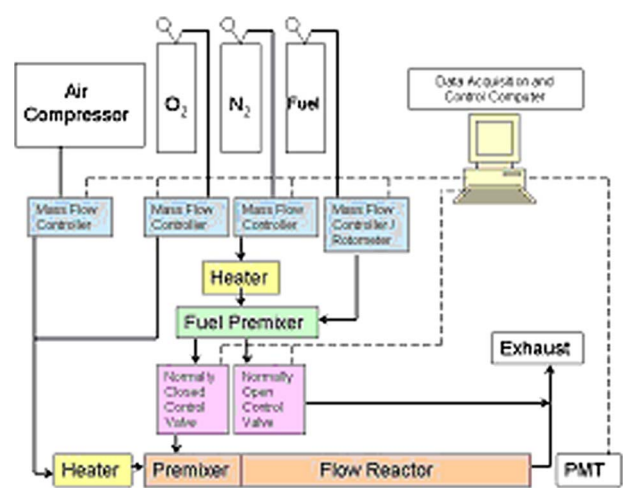

Fig. 3 Schematic of the experimental setup used for ignition delay time measurements

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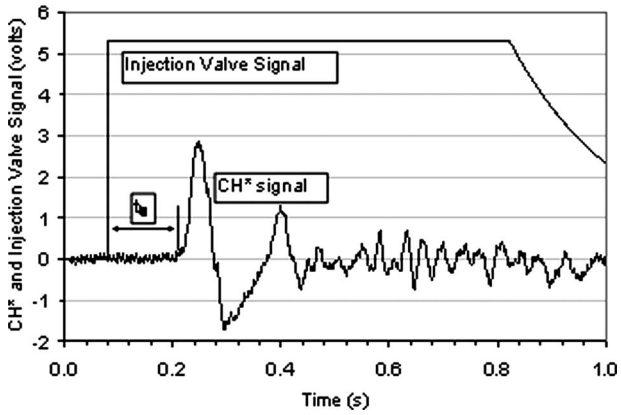

Fig. 4 Sample injection valve and CH* chemiluminescence signals and ignition delay time definition

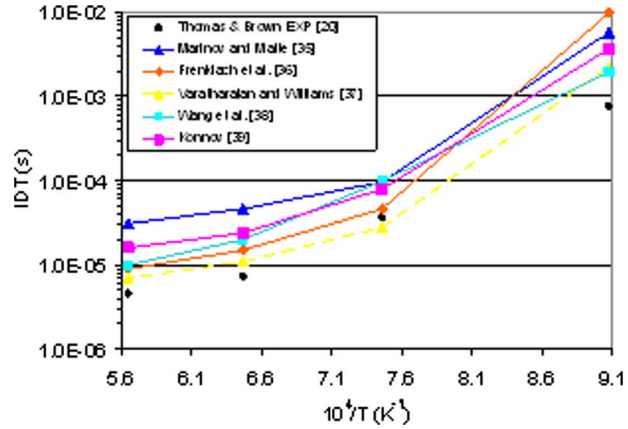

Fig. 5 Typical performance of available ethylene chemical kinetics mechanisms [35–39] in predicting experimental ethylene ignition data of Brown and Thomas [20] at temperatures between 1102 K and 1771 K and pressures between 1.4 atm and 3.4 atm

(PMT) was monitored and used to indicate the ignition of fuel-air mixture. The time difference between when the fuel was injected into the air stream and when the fuel and air mixture ignited was defined as the ignition delay time. A typical ignition event is shown in Fig. 4.

The difficulties of making autoignition delay time measurements in flow reactor apparatus include achieving "instantaneous" mixing of fuel and air, and a zero temperature gradient at the entrance to the test section to maintain ideal plug-flow conditions. In the current experiments, a swirler, followed by an expanding duct, was used to rapidly mix the fuel and air upstream of the flow reactor [27]. However, the mixing of the fuel and air still has a finite mixing time-scale. It was demonstrated that chemical and physical mixing effects upstream of the flow reactor can be accounted for by modeling the premixing section with a chemical reactor network of ideal reactors [29]. Therefore, in the present work, experiments were performed with ethylene to characterize the physical and chemical mixing effects in the premixing section on ignition delay time measurements.

A chemical reactor model (CRM), consisting of a large number of perfectly stirred reactors with infinitely small volumes in a series, was developed, based on the geometry and flow conditions in the transfer lines and premixing section to account for the mixing effects on the ignition delay time measurements. The mixing time upstream of the test section of the flow reactor was therefore numerically computed as the sum of the residence times of the series of perfectly stirred reactors representing the premixing section. The mixing time was then subtracted from the experimental measurements to obtain autoignition delay time. The computed mixing time-scale in the premixing zone ranged between 125 ms and 205 ms for various temperatures and flow rates studied.

The accuracy of several available chemical kinetics mechanisms in predicting existing ethylene combustion data was investigated in order to identify the most appropriate mechanism for use in the CRM. Because ethylene is a key intermediate in the oxidation of higher-order hydrocarbons, as well as an important fuel in high speed propulsion devices [19,30], combustion of this fuel was widely studied using a variety of techniques over a broad range of conditions. Ethylene autoignition delay times and decomposition profiles were reported in the literature [20,23,30–34]. The conditions of each experiment were modeled using existing chemical kinetics mechanisms [35–39], most of which were developed specifically to model ethylene combustion.

Figure 5 shows the typical performance of the available chemical kinetics mechanisms in predicting ethylene ignition. This figure shows the accuracy of the kinetics mechanisms in predicting autoignition delay times of ethylene reported by Brown and Thomas [20] at temperatures between 1102 K and 1771 K and pressures between 1.4 atm and 3.4 atm. The kinetics mechanism developed by Varatharajan and Williams [37], focusing on ethylene combustion, consistently predicted the experimental results of other researchers most accurately.

The mechanism developed by Varatharajan and Williams [37] was therefore used in the CRM to model the flow reactor apparatus in the current study. Mixing times calculated from the CRM were subtracted from the experimental data to obtain corrected autoignition delay times of ethylene. Figure 6 shows that these corrected autoignition delay times have good agreement with predictions made by the ethylene mechanism of Varatharajan and Williams [37]. Figure 6 also shows low-pressure (i.e., 1–5 atm) ethylene/air ignition delay time data from the literature [20,23,33] for comparison. Most of these experiments from the literature were conducted at higher pressures than atmospheric, resulting in autoignition delay times shorter than in the current study.

The resulting mixing times obtained in the ethylene experiments were then applied to the measurements made for all fuels in this study. While errors may be introduced in assuming the mixing times to be identical between fuels, the main factors contributing to the calculation of mixing times are bulk flow rates, temperature, and flow path geometry; all of which remain essentially constant throughout the experiment.

Atmospheric pressure autoignition delay time measurements were made for methane, ethane, propane, binary methane/ethane,

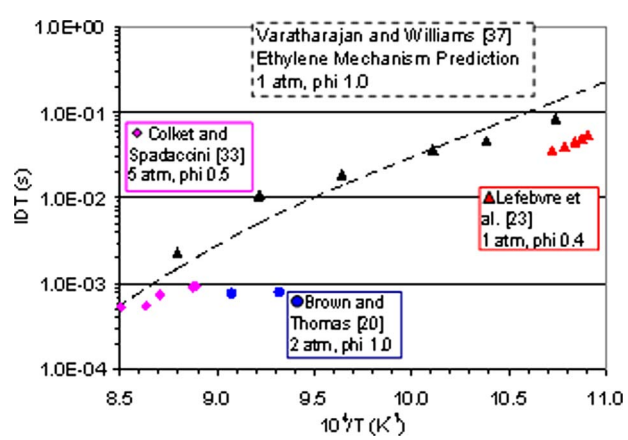

Fig. 6 Ethylene autoignition delay time measurements obtained in the present work (▲) compared with data from the literature [20,23,30,33] and predictions made by Varatharajan and Williams [37] on ethylene mechanism

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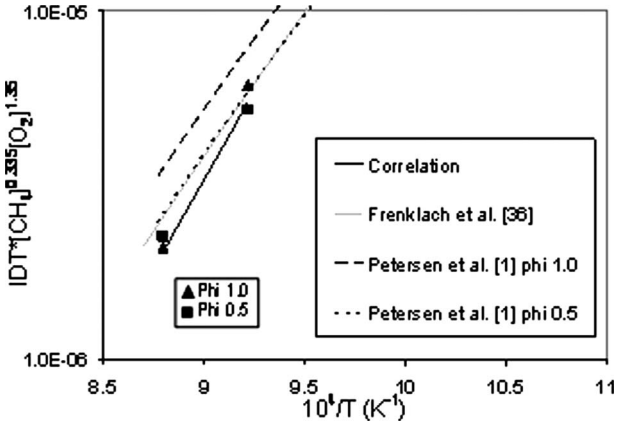

Fig. 7 Methane autoignition delay time measurements, Arrhenius correlation, GRI 3.0 predictions [36], and kinetics model predictions of Petersen et al. [1]

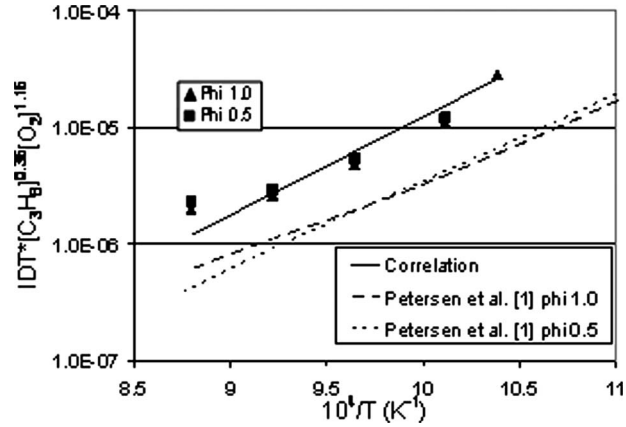

Fig. 9 Propane autoignition delay time measurements, Arrhenius correlation, and kinetics model predictions of Petersen et al. [1]

binary methane/propane, ternary methane/ethane/propane, and methane/ethane/CO_2 fuels at equivalence ratios ranging from 0.5 to 1.25 and temperatures ranging from 968 K to 1137 K. Detailed experimental results, including the fuel compositions and test conditions, as well as the experimental measurements of autoignition delay times can be found elsewhere [40]. The reproducibility of the ignition delay time measurements typically lies within 10% of the average autoignition delay time measurement at a given condition.

3 Autoignition Delay Time Measurements of Pure Fuels

Methane, ethane, and propane autoignition was studied at atmospheric conditions, equivalence ratios between 0.5 and 1.25, and temperatures ranging from 930 K to 1140 K. Figures 7–9 show the measured autoignition delay times of the pure fuels, as well as Arrhenius fits to the data of the following form:

$$\tau = A_f \exp(E_a/RT)[C_xH_y]^a[O_2]^b \tag{1}$$

where E_a is an effective global activation energy and A is an empirically determined constant. $[C_xH_y]$ and $[O_2]$ are molar concentrations of the fuel and oxygen, respectively. R is the universal gas constant and T is the initial temperature of the mixture in Kelvin. Regression analyses of the collected data at all T, $[C_xH_y]$, and $[O_2]$ were performed for each fuel to determine E_a, a, and b. Table 1 presents the derived Arrhenius correlations of autoignition

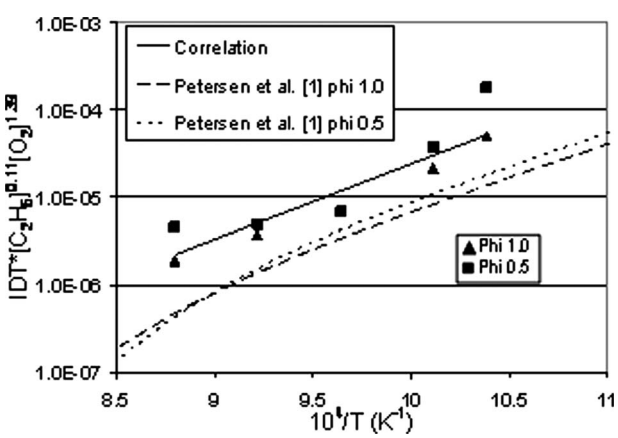

Fig. 8 Ethane autoignition delay time measurements, Arrhenius correlation, and kinetics model predictions of Petersen et al. [1]

delay time for pure methane, ethane, and propane, as well as methane/ethane and methane/propane mixtures. The predictions of methane autoignition delay time using the GRI 3.0 chemical mechanism [36], which was well-validated for methane combustion, are presented as well to show the accuracy in modeling the results reported herein. Predictions of autoignition delay time using the detailed C1-C3 kinetics mechanism developed by Petersen et al. [1] are presented for all fuels in the current study.

Figure 7 shows ignition delay time data for methane from the present work. Because only two test temperatures resulted in successful autoignition events for the methane experiments, it is difficult to draw any meaningful conclusions about the resulting rate parameters. E_a for methane oxidation in the temperature regime studied was found to be 46.6 kcal/mol. Studies available in the literature at various temperature and pressure regimes reported activation energies ranging from 19 kcal/mol to 51.8 kcal/mol [5,11,13,15,23,41].

Figure 7 compares the methane ignition delay time data obtained in the current work with the detailed kinetics model predictions. The predictions made by the GRI 3.0 mechanism have reasonable agreement with the experimental results. The kinetics mechanism of Petersen et al. [1] predicted the measurements fairly well at 0.5 equivalence ratio, but overpredicted the measured autoignition delay times of the stoichiometric mixture.

Similar autoignition delay time measurements were made for pure ethane and propane fuels. Figures 8 and 9 show the ignition delay time data for ethane and propane, respectively, compared with Arrhenius correlations, as well as detailed kinetics model predictions of Petersen et al. [1]. Figure 10 compares the propane/air ignition delay time data of the current work with the data from the literature obtained at 1 atm [42,43]. The detailed kinetics mechanism of Petersen et al. [1] predicts a lower ignition delay time than the experimental data below 1100 K, while the model agrees with the experimental data of Kim et al. [43] fairly well, as shown in Fig. 10.

The regression analysis, which is plotted with the data in Figs. 8 and 9, computed E_a for ethane (40.0 kcal/mol) slightly larger than for propane (38.5 kcal/mol). The H-atom abstraction from secondary C-H sites requires less energy than from primary sites. Therefore, ethane and propane fuels provide lower E_a for ignition than methane, which solely consists of primary C-H sites. deVries et al. [18] studied ethane ignition in the temperature range of 1218 K to 1860 K at pressures of 0.57 atm to 3 atm and for equivalence ratios ranging from 0.5 to 2. The activation energy reported by deVries et al. [18] for ethane oxidation was 39.6 kcal/mol, which agrees well with the current study. For propane ignition, Cadman et al. [21] reports an activation energy of 9.1 kcal/mol at 5 atm,

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Table 1 Derived Arrhenius expressions for autoignition delay time for pure fuels methane, ethane, and propane, and binary methane/ethane and methane/propane mixtures

Fuel constituent(s)	Arrhenius autoignition delay time correlation
Methane	$\tau_{ig}(s)=2.23^{*}10^{-15}[CH_4]^{-0.335}[O_2]^{-1.35}\exp(46.6/RT)$
Ethane	$\tau_{ig}(s)=6.9^{*}10^{-14}[C_2H_6]^{-0.11}[O_2]^{-1.39}\exp(40.0/RT)$
Propane	$\tau_{ig}(s)=4.8^{*}10^{-14}[C_3H_8]^{-0.36}[O_2]^{-1.16}\exp(38.5/RT)$
Methane/ethane	$\tau_{ig}(s)=7.1^{*}10^{-17}[CH_4]^{-0.41}[C_2H_6]^{-0.15}[O_2]^{-1.8}\exp(42.2/RT)$
Methane/propane	$\tau_{ig}(s)=6.1^{*}10^{-17}[CH_4]^{-0.55}[C_3H_8]^{-0.35}[O_2]^{-1.3}\exp(41.8/RT)$

$\phi=1$ with T from 850 K to 1280 K, and an activation energy of 25.8 kcal/mol for pressures of 5 atm, $\phi=0.5$ with T from 998 K to 1357 K.

The ethane and propane measurements shown in Figs. 8 and 9 suggest a decrease in E_a at $T>1000$ K. This trend is most prominent for ethane at the leaner equivalence ratios. Other researchers have noted this trend at lower temperatures (near 800 K) for elevated pressures (20 atm) and it has been associated with a negative temperature coefficient for hydrocarbon ignition [16]. The predictions of autoignition delay time by the kinetics mechanism of Petersen et al. [1] do not capture this reduction in effective E_a but rather predict an effectively constant E_a over the entire range of temperatures studied, as indicated in Figs. 8 and 9. Further, the mechanism underpredicts the experimental measurements of autoignition delay time for both ethane and propane.

The autoignition delay times measured for pure ethane and propane show a slight decrease when increasing the equivalence ratio from 0.5 to 1.0. The predictions of Petersen et al. [1] suggest that this trend is followed only below a certain temperature, and that as temperature increases well above 1000 K, increasing equivalence ratio will increase autoignition delay time. This predicted trend at high temperatures was not observed in the experiments.

This shifting trend with temperature regime was experimentally verified by Healy et al. [25], who studied methane/propane fuel mixtures. The authors have observed that fuel-lean mixtures react faster than fuel-rich mixtures at high temperatures, and fuel-rich mixtures react faster than fuel-lean mixtures at lower temperatures. The authors attribute this phenomenon to the fact that at lower temperatures, chain-branching is dependant on fuel concentration. Therefore fuel-rich mixtures will exhibit faster kinetics, resulting in quicker autoignition. At higher temperatures, the $H+O_2\rightarrow O+OH$ reaction becomes the rate-limiting chain-branching reaction, and thus, reactivity is more strongly dependent on $[O_2]$ than on $[C_xH_y]$. Therefore, mixtures with lower ϕ and therefore higher $[O_2]$ will have shorter autoignition delay time.

The temperature at which the shift in controlling kinetics occurs has been shown to be highly dependent on pressure. Healy et al. [25] found very good agreement of their data, which was collected at pressures ranging from 10 atm to 30 atm, with predictions made using the kinetics mechanism of Petersen et al. [1]. The authors found that as pressure increases, the critical temperature for the shift in controlling kinetics increases. The atmospheric pressure measurements made in the current study, however, do not agree well with predictions made by this kinetics mechanism. The kinetics mechanism of Petersen et al. [1] predicts the critical temperature for a shift in the controlling kinetics at 1112 K for ethane and 1025 K for propane at atmospheric pressure. For all temperature conditions experimentally studied for methane, ethane, and propane autoignition, it was found that increasing ϕ reduced autoignition delay time, indicating that the experiments were performed at temperatures below the critical temperature associated with the kinetics regime shift at atmospheric pressure. The measurements made in the current study can therefore provide critical validation data for further development of the kinetics mechanisms at atmospheric pressure.

4 Autoignition Delay Time Measurements of Binary Methane-Based Fuel Mixtures

Autoignition delay times were measured for binary mixtures of methane/ethane and methane/propane in air. The following binary compositions were investigated at equivalence ratios of 0.5 and 1.0: 95/5, 90/10, 75/25, 50/50, and 25/75. Figures 11 and 12 show the measured autoignition delay times of the binary methane fuel blends, as well as Arrhenius fits to the data of the form shown in Eq. (2). The mole fractions of methane and ethane used to generate the curve using the Arrhenius correlation in Fig. 11 were 0.0801 and 0.0089, respectively. The mole fractions of methane

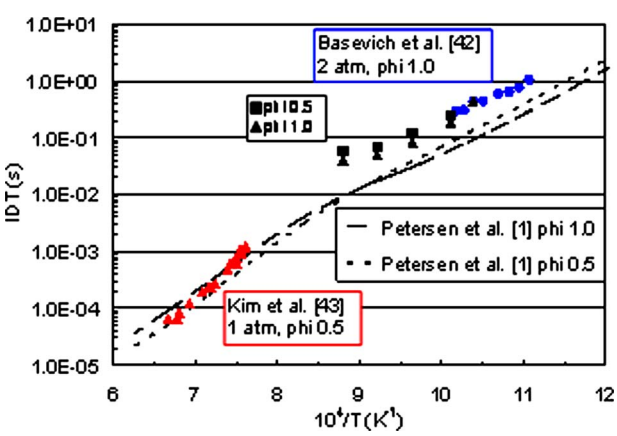

Fig. 10 Propane autoignition delay time measurements obtained in the present work ($\phi=0.5$■, $\phi=1.0$▲) compared with atmospheric pressure measurements reported in the literature [42,43] and kinetics model predictions made by Petersen et al. [1]

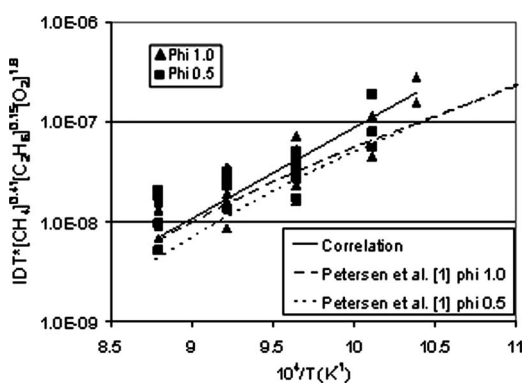

Fig. 11 Methane/ethane autoignition delay time measurements, Arrhenius correlation, and Petersen et al. [1] mechanism predictions. The mole fractions of methane and ethane used in the Arrhenius correlation were 0.0801 and 0.0083, respectively.

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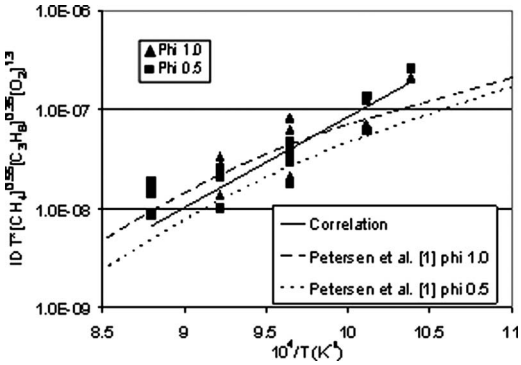

Fig. 12 Methane/propane autoignition delay time measurements, Arrhenius correlation, and mechanism predictions of Petersen et al. [1]. The mole fractions of methane and propane used in the Arrhenius correlation were 0.0393 and 0.0044, respectively.

and propane used to generate the curve using the Arrhenius correlation in Fig. 12 were 0.0393 and 0.0044, respectively

$$\tau = A_f \exp(E_a/RT)[\mathrm{CH_4}]^{a1}[\mathrm{C_xH_y}]^{a2}[\mathrm{O_2}]^b \tag{2}$$

Several researchers derived Arrhenius expressions for methane oxidation, but only Spadacccini and Colket [7] attempted to account for the effects of hydrocarbon additives in the methane-based fuel mixtures. In their correlation, all non-methane hydrocarbons are lumped together in one term. In the current study, Arrhenius fits are made for both methane/ethane and methane/propane binary fuel mixture oxidation. Figures 11 and 12 also compare the data to predictions by the kinetics mechanism of Petersen et al. [1] for fuel mixtures composed of 90% methane and 10% ethane or propane.

Figure 13 shows how increasing the higher hydrocarbon concentration in a methane-based fuel blend impacts autoignition delay time. Methane/propane binary mixture autoignition delay time is presented as a function of propane mole fraction in the fuel. It can be noted that with less than 10 mol % addition of ethane or propane to methane, methane autoignition delay times can be reduced significantly. This trend can be seen most clearly in the data obtained at 1085 K. A conclusion cannot be made regarding the effect of small additions of ethane or propane to methane at the lower temperatures because autoignition was not achievable in the experimental reactor for the residence times of this study. For the higher test temperatures, reductions in methane autoignition delay time of up to 50% were observed in the mixtures composed of up to 10% ethane or propane by volume. On the other hand, promotional effects of ethane and propane on methane autoignition greatly diminish above 20% additions.

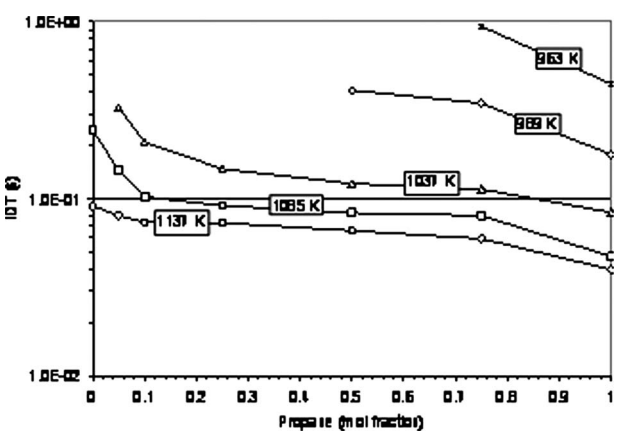

Fig. 13 Methane/propane autoignition delay time as a function of propane mole fraction in the fuel, ϕ=1.0

Figure 13 also shows the significant increase in autoignition delay time of pure propane with the addition of methane. This trend was also seen in the addition of methane to pure ethane. Increases in autoignition delay times of ethane and propane of up to 50% were observed with the addition of methane, making the resulting concentration of the methane blend 25% methane/75% ethane or propane.

Differences in autoignition delay times were observed if ethane or propane was an additive; ethane addition tended to reduce autoignition delay times of methane blends more than propane. The majority of tests indicate that ethane addition typically reduced methane autoignition delay time about 10% more than that of propane addition. While some results show this difference as high as 40%, few results show similar reductions with either ethane or propane, and even some results show larger reductions with propane than ethane.

Very few researchers have derived global rate parameters for the ignition of the fuel blends studied. Only Petersen et al. [5] derived global activation energies for the oxidation of the fuel blends. In their study, an 80% methane/20% propane mixture was found to have an effective activation energy of 41.9 kcal/mol between 1189 K and 1615 K at 12.2 atm. This agrees with E_a =41.8 kcal/mol, derived from the data collected in the current study. Petersen et al. [5] also found an effective activation energy of 42.1 kcal/mol for 90% methane/10% ethane mixture oxidation, which agrees with E_a=42.2 kcal/mol, derived from the data collected in the current study. The derived E_a for the methane fuel blends was found to be lower than E_a for pure methane, but higher than E_a for ethane or propane pure fuel additive.

The kinetics mechanism of Petersen et al. [1] predicted a very similar autoignition delay time for the methane/ethane and methane/propane blends. The mechanism overpredicted the autoignition delay times of methane, but underpredicted the autoignition delay times of pure ethane and propane. However, the mechanism performed very well in predicting autoignition delay times of methane blends containing low concentrations of ethane and propane. The mechanism of Petersen et al. [1] predicts a reduction in autoignition delay time with a reduction in equivalence ratio for the conditions of the binary mixtures studied. This trend was again evident in the predictions made above a critical temperature for pure ethane and propane autoignition.

The kinetics mechanism predicts a shift in the controlling kinetics of methane/ethane fuel blend oxidation near 930 K. This prediction was not verified by the experiments, as fuel-rich methane/ethane mixtures were found to have shorter autoignition delay times than fuel-lean mixtures for the entire temperature range investigated. A similar trend was predicted for methane/propane mixtures. Figure 12 indicates that, for the conditions studied, the mechanism predicts a reduction in methane/propane mixture autoignition delay time with decreasing equivalence ratio. It was found in the current study that the critical temperature for a shift in the controlling kinetics, indicated by the effect of equivalence ratio on ignition delay time, was lower than the predictions by this mechanism. The data collected in the current study will provide a basis for future kinetics model validation in the atmospheric pressure regime.

5 Autoignition Delay Time Measurements of Ternary Methane-Based Fuel Mixtures

Methane-based ternary blends containing both ethane and propane were studied to investigate the effect of multiple hydrocarbon addition on methane autoignition delay time to provide further validation data for natural gas chemical kinetics mechanisms. Two ternary fuel mixtures were studied at ϕ=0.5 and 1.0: mixture 1: 50% methane, 25% ethane, and 25% propane; and mixture 2: 85% methane, 10% ethane, and 5% propane. Figure 14 shows the

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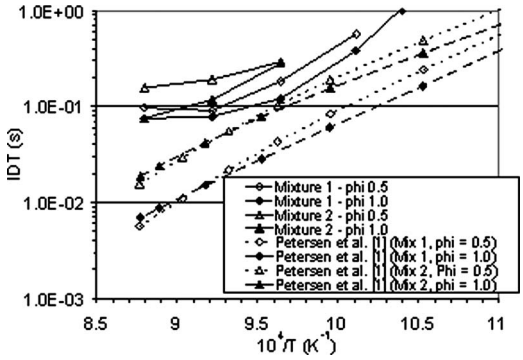

Fig. 14 Ternary methane/ethane/propane autoignition delay time measurements and mechanism predictions of Petersen et al. [1]. Mixture 1: 50% CH_4, 25% C_2H_6, and 25% C_3H_8; mixture 2: 85% CH_4, 10% C_2H_6, and 5% C_3H_8.

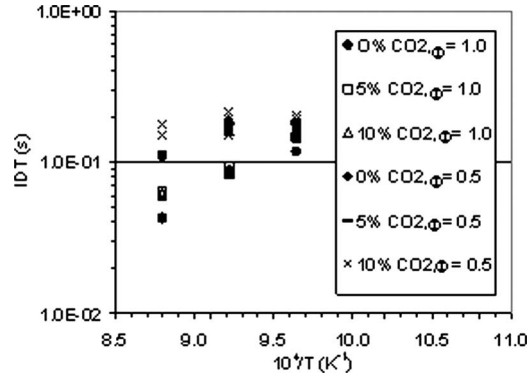

Fig. 15 Effect of CO_2 addition on 75% methane/25% ethane fuel blend autoignition delay time. CO_2 composition is reported as mol % in the fuel mixture.

measured autoignition delay times for these mixtures, as well as predictions of autoignition delay time made by the kinetics mechanism of Petersen et al. [1]. As with the pure fuels and binary methane fuel blends, autoignition delay times of the ternary methane fuel blends exhibited the same trends with temperature and equivalence ratio. Figure 14 shows the same near exponential decrease in autoignition delay time with increased temperature. And again, this reduction in autoignition delay time decreases at higher temperature, again, more evidently for leaner mixtures.

Autoignition delay times of mixture 1 were significantly (up to 50%) shorter than those of mixture 2 at the same conditions. This was expected as mixture 1 had significantly more ethane and propane added to the methane than in mixture 2. Addition of both ethane and propane had been observed to decrease the autoignition delay time of methane. This data will serve to improve the accuracy of chemical kinetics tools in predicting typical natural gas ignition behavior.

In comparing the autoignition delay times of binary mixtures made up of 50% methane and ternary mixture 1 (50% methane/25% ethane/25% propane), the values of autoignition delay times for ternary mixture 1 fall between the autoignition delay times of a binary methane/ethane mixture and a binary methane/propane mixture. While the autoignition delay times measured for all three of these mixtures are within about 10% of one another for each test temperature, the results indicate that the promotional effects of ethane and propane on methane autoignition are individually significant.

Conversely to what was seen in the binary methane/ethane fuel blends, the kinetics mechanism of Petersen et al. [1] underpredicted the autoignition delay times measured for the ternary methane-based fuels. This indicates that perhaps the mechanism overemphasizes the synergetic effect of natural gas surrogate components on reducing methane autoignition delay time at atmospheric pressure. As was seen in the autoignition delay time measurements of the pure fuels, and to a lesser extent, the binary methane blends, a very significant asymptotic trend of autoignition delay time reduction was evident at the higher temperatures for the ternary fuel blends. This trend was not captured by the mechanism predictions.

The temperature at which the shift in controlling kinetics occurs was predicted to be 1100 K for mixture 1 and 1075 K for mixture 2 by the mechanism developed by Petersen et al. [1]. As was found for the pure fuel and binary methane mixtures studied, autoignition delay times decreased with increasing equivalence ratio for the range of temperatures studied for the ternary mixtures as well. Therefore, again, it appears that the kinetics mechanism underpredicts this critical temperature for the mixtures and conditions studied.

6 Autoignition Delay Time Measurements of Methane/Ethane/CO_2 Fuel Mixtures

The effect of CO_2 addition on the ignition of natural gas type fuel blends was investigated. One binary methane/ethane fuel mixture was selected to serve as the basis for these tests: 75% methane/25% ethane. This mixture was then diluted with CO_2, such that mixtures containing 5% and 10% CO_2 by volume were created. Autoignition delay times were measured for both stoichiometric mixtures and mixtures having an equivalence ratio of 0.5. The autoignition delay times measured for the mixtures containing CO_2 are compared with the measurements made for the original 75% methane/25% ethane mixtures in Fig. 15.

The resulting autoignition delay times for the mixtures containing CO_2 do not significantly deviate from the measurements of autoignition delay times for the mixture containing no CO_2. Only one data series offered insight into a possible effect of CO_2 on autoignition delay time. For the mixtures at an equivalence ratio of 0.5, it was observed that at the highest test temperature (1137 K), adding 5% CO_2 to the mixture increased the autoignition delay time by only 2%, but increasing the concentration to 10% CO_2 increased the autoignition delay time an additional 46%. This observation can be explained in part, due to the third-body collision efficiencies of CO_2 being an order of magnitude greater than those of N_2 [36]. Therefore, the presence of CO_2 promotes recombination reactions, and hence, reduces the fuel reactivity. Due to the relatively small amount of data collected in this investigation, however, further experiments are needed to quantify the effect of CO_2 on fuel mixture ignition delay time.

7 Conclusions

Autoignition delay time experiments were conducted in an atmospheric flow reactor for methane, ethane, and propane pure fuels; binary methane/ethane and methane/propane; ternary methane/ethane/propane; and methane/ethane/CO_2 fuel mixtures. The measurements provide a comprehensive database, which can be used to improve the accuracy of chemical kinetics mechanisms for predicting ignition of typical natural gas fuel blends. Most experimental autoignition delay time data reported in the literature for the fuels studied herein has been collected at elevated pressures and temperatures using shock tubes and rapid compression machines. The measurements made at low to intermediate temperature and atmospheric pressure fill some regions in the parameter space of the published autoignition delay time of natural gas fuels where previous data was sparse. Important trends of reductions in autoignition delay times of the fuels and fuel blends with increasing temperature and equivalence ratio were observed. Fur-

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ther, a reduction in the decrease of autoignition delay times at higher temperatures was observed for the pure fuels and fuel blends studied.

Arrhenius rate expressions providing global activation energies for the oxidation of the pure alkanes and methane fuel blends have been reported. Reductions in activation energy of methane oxidation were observed with the addition of both ethane and propane. Further, the trends observed in this study indicate that addition of ethane and propane to methane has a very significant promotional effect on methane autoignition, as found by other researchers [9,24]. The relative reactivity of methane, ethane, and propane was determined, and global activation energies of the oxidation of these pure fuels and mixtures of these fuels were derived. For the conditions studied, methane oxidation activation energy was found to be 46.6 kcal/mol. It was found that both ethane and propane have very similar oxidation activation energies (40.0 kcal/mol and 38.5 kcal/mol, respectively). As additives to methane, both ethane and propane have the effect of reducing oxidation activation energy. Binary methane/ethane fuels were found to have oxidation activation energy of 42.2 kcal/mol and binary methane/propane fuels of 41.8 kcal/mol. Differences in autoignition delay times were observed for ethane versus propane additives; ethane addition tended to reduce autoignition delay times of methane blends about 10% further than propane.

An important finding, because of its application to practical natural gas fuels, was that the most dramatic decrease (30–50%) in autoignition delay time of binary, methane-based fuels occurs with the addition of 5–10% of ethane or propane. Further addition of ethane or propane continued to reduce autoignition delay time, but not as dramatically. This trend was found for all binary mixtures of methane/ethane and methane/propane.

This study indicates that current kinetics predictive tools seem to capture the promotional effects of ethane and propane individual additives on methane autoignition, but not the effect of combinations of ethane and propane on methane autoignition delay time adequately. Further, the data collected indicate that for the temperature range studied (930–1140 K), autoignition delay times of methane, ethane, and propane pure fuels and mixtures decrease with increasing equivalence ratio at atmospheric pressure. This is not captured by the current kinetics mechanisms, which predict a shift in the controlling oxidation kinetics at atmospheric pressure in the temperature range studied. Additional mechanism development, inclusive of the data collected herein, is needed to better model oxidation of natural gas fuel blends. Interesting trends of increased autoignition delay time of methane-based fuel blends were seen with CO_2 addition; however, because such little data was collected in this area, further study is required to draw significant conclusions about the effect of a fuel's CO_2 concentration on its autoignition delay time.

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Nomenclature

Φ = equivalence ratio
τ_{ig} = autoignition delay time
a = exponential fuel dependency
b = exponential oxygen dependency
A = preexponential factor
E_a = effective activation energy
IDT = autoignition delay time
R = universal gas constant (1.987 cal mol^{-1} K^{-1})

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